

are specified in gauges. Solder of 18 Standard Wire Gauge (SWG) is used for general purposes. Thinner soldering wire of 22 SWG or 24 SWG is used for printed circuit assembly.

Flux

Perfect solder joints can be accomplished only with properly cleaned soldering surfaces. The flux present in multicore solders is a chemical for removing the oxide film deposited on the surfaces to be joined. The corrosive action of flux at melting temperatures removes metal oxides very fast. It also prevents reformation of new oxides, which allows the solder to form a rigid bond. Flux helps molten solder flow more easily over the joint.

Flux melts at a temperature lower than the solder temperature. Thus it performs the task of cleaning the surfaces before soldering takes place. A variety of flux are available for many purposes and applications. The most common types specified by rosin content are 'no clean', rosin mildly activated, rosin activated and water soluble. Flux consists of natural or synthetic rosins. Flux used in hand soldering are either pure rosins or rosins combined with mild activators.

Key to good soldering

Cleaning of the surface, soldering at the right temperature and for the right duration, and adequate solder coverage are the key factors affecting the quality of solder joints.

It is absolutely necessary to ensure that the leads of components are free from grease, oxidation and other contamination. The oxide layer repels molten solder. It is very difficult to solder old components or tracks on the PCB because of the layer of oxidation that builds up on the surface of the component leads. Even surfaces that look clean may have a thin, invisible film of oxide on them. So for appropriate

solder bonding, surfaces must be cleaned using flux.

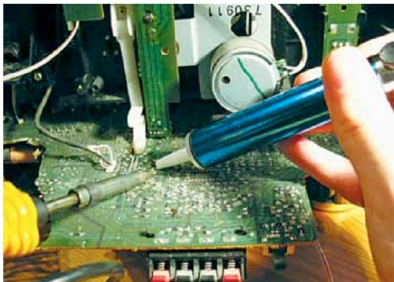
It is important that the surfaces to be soldered together are brought to the same temperature. When both are heated at a high temperature, solder flows evenly to make a good mechanical and electrical joint.

The parts to be joined should be heated with a soldering bit for the right time length. Excessive heating may damage the components or the printed circuit track. The heating period depends on the size of the joint and the temperature of the soldering bit. Semiconductor components such as diodes, transistors and ICs are sensitive to heat and should not be overheated.

Soldering: the process

When the soldering iron bit is adequately hot, apply a little solder on the flattened tip and wipe it off with a piece of damp cloth or sponge. The solder will form a thin layer on the bit. This process is known as 'tinning' the bit.

Heat up the joint with the bit and continue heating while applying solder. The molten solder flows quickly from the bit onto both parts to be joined. It is important to use the right amount of solder. Remove the iron and allow the joint to cool. During the cooling period, if the joint is disturbed, it may become dry and create serious problems in the working of the circuit.



Desoldering using a vacuum plunger (right) and a soldering iron (left) (Courtesy: Wikimedia Commons)

Sometimes it becomes very difficult to trace out this defect.

Post-solder flux cleaning

After the soldering process is complete, remove the residual flux from the PCB. Clean it by dissolving it in a solvent and wiping the area dry with a tissue paper. The complete assembly should be totally free from flux and residual cleaning agents. The cleaning method and solvent solution used should not

have any adverse effect on the components and the connections.

Wetting action

While soldering, if molten solder comes into contact with a copper surface, the solder dissolves and penetrates the copper surface. Solder and copper form a new alloy. This solvent action is called 'wetting' and forms an inter-metallic bond between the two parts.

Wetting can occur only if the surface of copper is free of contamination and oxide film, and the solder and the work surface have reached the proper temperature.

Resoldering and desoldering

Resoldering. It is very difficult to correct poorly made joints. So make good joints in the first place. Try to prevent re-soldering of joints as much as possible. A dry or disturbed solder joint usually requires reheating and re-flow of solder by applying a suitable quantity of flux.

Desoldering. Desoldering is required when components in the circuit need to be removed or replaced. To remove a damaged or defective component, either a desoldering pump or a desolder braid can be used.

Using a desoldering pump. The desoldering pump consists of a spring-loaded plunger. During the desolder-